

PRACTICAL EXAMINATION

Subject code: DBI202

Duration: 85 minutes

Requirements:

- + Download the file **PE_DBI202_.sql** and execute it to create the database and the tables that are already in the file.
- + Students must complete their exams in 2 files: file named **PE_DBI202_.sql** for SQL code and word file **PE_DBI202.docx** containing the **ERD** for question 1.

1. (2 points) In the picture are two entities extracted from a simplified ERD of a bank. Complete the ERD according to the given details:

- Each client can have many accounts.
- Each account is of one account type.

Each account has an ID and a balance, which is the amount of money left in the account.

(Add to the existing diagram in the exam).

Use the diagram below to complete question 1 (you can draw it in MS Word).



Entities Description

- **Client:** Contains personal information about each client, including Id (of the client), FirstName, LastName.
- **AccountType:** Stores information about each type of account, including Id (of the type) and Name.
- **Account:** Contains information about each account, including account Id and account balance.

2. (2 points) **From the complete ERD in question 1, write SQL statements to create the table(s)** corresponding to the entity(ies) added to the diagram. Insert **5 appropriate data rows** into each newly created table.

Notice: The balance of an account has up to 15 digits including 2 digits after the decimal point and has a default value of 0. Suppose that the currency is USD.

3. (2 points) Write a **stored procedure** to withdraw money from an account. In order for a withdrawal to be successful, the amount to be withdrawn must be a positive number, the account ID must be valid and the balance of the account must not be lower than the amount to be withdrawn. In case any criterion above is violated, rollback and/or raise an error.

4. (2 points) **Create a table** – NotificationEmails (Id, Recipient, Subject, Content). **Add**

2. (2 points) **From the complete ERD in question 1, write SQL statements to create the table(s)** corresponding to the entity(ies) added to the diagram. Insert **5 appropriate data rows** into each newly created table.

Notice: The balance of an account has up to 15 digits including 2 digits after the decimal point and has a default value of 0. Suppose that the currency is USD.

3. (2 points) Write a **stored procedure** to withdraw money from an account. In order for a withdrawal to be successful, the amount to be withdrawn must be a positive number, the account ID must be valid and the balance of the account must not be lower than the amount to be withdrawn. In case any criterion above is violated, rollback and/or raise an error.

4. (2 points) **Create a table** – NotificationEmails (Id, Recipient, Subject, Content). **Add a trigger** to the table containing information about accounts and create a new message whenever this accounts table is updated. The following data is required to be filled for each email:

- Recipient – AccountId
- Subject – “Update on account: {AccountId}”
- Content - “On {date} your balance was changed from {old balance} to {new balance}.”

For example:

Id	Recipient	Subject	Content
1	1	Update on account: 1	On 05-11-2023 14:09 your balance was changed from 113.12 to 103.02.
...	...	...	...

5. (2 points) **Create a view** that displays the first names and last names of all clients who have **more than \$300 in total** of all their accounts.

End